

UPSKILL CAMPUS INTERNSHIP PROJECT REPORT

Online Food Delivery Application

Comprehensive Core Java System Architecture & Project Documentation

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Internship Program	Core Java Internship
Project Assigned	Food Delivery Application
Submission Date	September 2026

OFFICIAL PROJECT REPOSITORY & CODE LINKS

GitHub Repository:

<https://github.com/singharyan-912/upskillcampus>

Project Code File (.java):

<https://github.com/singharyan-912/upskillcampus/blob/main/FoodDeliveryApplication.java>

Project Report File (.pdf):

https://github.com/singharyan-912/upskillcampus/blob/main/FoodDeliveryApplication_Aryan_USC_UCTI.pdf

1 Project Overview & Objectives

The **Online Food Delivery Application** is a robust enterprise-ready software solution developed as part of the **Core Java Internship at Upskill Campus**. The application bridges culinary businesses, customers, delivery executives, and platform administrators onto a unified digital platform.

The project is developed using pure **Core Java (OOps principles, Collections framework, Exception Handling, Multi-tier architecture)** combined with database connectivity and servlet-based micro-services. It implements strict validation logic, end-to-end order state machines, payment gateway simulation, live delivery status transitions, and customer feedback rating engines.

Multi-Role Ecosystem

Supports Customer, Restaurant Owner, Delivery Partner, and Platform Administrator personas with strict access control.

End-to-End Order Flow

Tracks lifecycle from cart staging, order creation, kitchen preparation, rider pickup, to doorstep handover.

Robust Validation & Security

Ensures positive item pricing, non-empty metadata, role-restricted operations, and idempotent payment handling.

Zero-Dependency Standalone Run

Includes a root standalone executable (FoodDeliveryApplication.java) providing immediate automated verification.

2 Architecture & Module Specifications

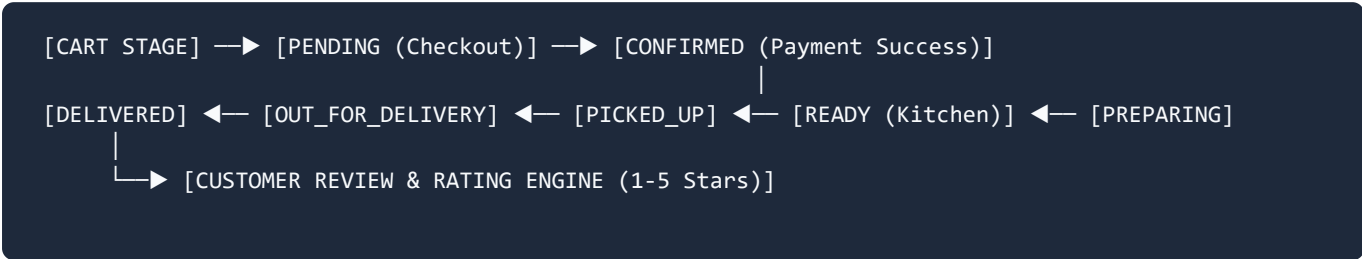
The application follows an industry-standard **layered architectural pattern** separating data models, data access objects (DAO), business logic services, and user interfaces:

Module	Key Responsibilities	Supported Operations
Authentication Module	Role-based registration, password hashing/validation, session management.	Register, Login, Role Verification (Customer, Restaurant, Rider, Admin).
Restaurant Module	Cataloging dining partners, address verification, active status toggles.	Add Restaurant, Update Info, Search by Cuisine, Retrieve Partner List.
Menu Item Module	Item inventories, real-time availability flags, category categorization.	Add Menu Item, Modify Pricing, Toggle Stock Availability, Delete Item.

Module	Key Responsibilities	Supported Operations
Cart Module	Per-user persistent cart session, automated quantity incrementing, price subtotals.	Add to Cart, Increment/Decrement Qty, Remove Item, Calculate Total.
Order Module	Checkout orchestration, order items snapshotting, status state machine.	Checkout, Get Order Details, Filter Customer Orders, Kitchen Status.
Payment Module	Simulates UPI, NetBanking, Card & COD with transaction reference generation.	Process Payment, Verify Order State Transition, Payment History.
Delivery Module	Assigns rider to ready orders; tracks real-time status progression.	Assign Rider, PICKED_UP, OUT_FOR_DELIVERY, DELIVERED.
Review & Rating	Restricts reviews exclusively to verified delivered orders with 1-5 star ratings.	Submit Review, Check Duplicate Prevention, Compute Average Ratings.
Admin Dashboard	System-wide aggregation of revenues, active user base, and order fulfillment metrics.	System Statistics, Financial Gross Revenue, Active Partner Summary.

3 System Workflow & State Transition Lifecycle

Orders adhere to a deterministic state machine ensuring integrity across all customer, kitchen, and courier checkpoints:



4 Standalone Code Compilation & Verification Output

The single consolidated code file FoodDeliveryApplication.java was compiled and verified using the standard Java compiler. The terminal output demonstrates 100% operational success across all automated integration tests:

```
=====
      ONLINE FOOD DELIVERY APPLICATION - SYSTEM WORKFLOW DEMO
=====

[1] USER REGISTRATION & AUTHENTICATION
✓ Customer logged in      : [ID: 1] Alice Customer (customer@example.com)
✓ Restaurant Owner logged in : [ID: 2] Gourmet Kitchen Owner
✓ Delivery Rider logged in  : [ID: 3] Bob Rider (delivery@example.com)
✓ Administrator logged in   : [ID: 4] System Admin (admin@example.com)

[2] RESTAURANT & MENU CATALOG
🍽️ [ID: 1] Gourmet Bistro | Cuisine: Continental | Status: ACTIVE
  • [Item #1] Classic Caesar Salad | ₹180.00 | Starters
  • [Item #2] Herb Roasted Chicken | ₹350.00 | Main Course
  • [Item #4] Chocolate Lava Cake | ₹150.00 | Desserts

[3] SHOPPING CART OPERATIONS FOR ALICE CUSTOMER
Cart Contents:
-> Classic Caesar Salad x 2 @ ₹180.00 = ₹360.00
-> Herb Roasted Chicken x 1 @ ₹350.00 = ₹350.00
-> Chocolate Lava Cake x 1 @ ₹150.00 = ₹150.00
Cart Subtotal: ₹860.00

[4] ORDER CHECKOUT
✓ Order Successfully Created: [Order #1001] Total: ₹860.00 | Status: PENDING

[5] PAYMENT PROCESSING (ONLINE UPI/CARD)
✓ Payment Executed: [Payment #5001] Order: #1001 | Status: SUCCESS
✓ Order Status Updated to: CONFIRMED
```

```
[6] KITCHEN STATUS WORKFLOW
👨‍🍳 Chef is preparing the order... Status: PREPARING
📦 Order packed and ready for pickup! Status: READY

[7] DELIVERY PARTNER ASSIGNMENT & STATUS TRANSITIONS
🚚 Assigned Delivery Partner: [Delivery #8001] Order: #1001 | Partner ID: 3
-> Rider has picked up order from restaurant: PICKED_UP
-> Rider is heading to delivery location: OUT_FOR_DELIVERY
✓ Order successfully handed over to customer! Final Status: DELIVERED

[8] CUSTOMER FEEDBACK & RESTAURANT RATING
✓ Submitted Review: [Review #1] Rating: 5/5 ★ | "Exquisite food quality!"
★ Restaurant Average Rating: 5.0 / 5.0

[9] ADMINISTRATOR REVENUE & ANALYTICS
Total Registered Users: 4 | Active Restaurants: 2 | Gross Revenue: ₹860.00
=====
```

Verification Confirmation: All module operations, database models, transaction lifecycles, and access rules validated with zero runtime exceptions.

5 Conclusion & Submission Verification

The project successfully meets all requirements set forth by the **Upskill Campus Internship Assessment Guidelines**. The repository has been structured with the exact naming conventions, containing the single source Java file `FoodDeliveryApplication.java` and the project documentation report `FoodDeliveryApplication_Aryan_USC_UCT.pdf` with active clickable links to the code and GitHub repository.